MIL-G-82253C
15 May 1978
SUPERSEDING
MIL-G-82253B
6 October 1972

MILITARY SPECIFICATION

# GLOVES, CLOTH, VINYL DIPPED, GENERAL PURPOSE

## (WATER, AVIATION FUEL, HYDROGEN PEROXIDE, OILS)

This specification is approved for use by the Naval Supply Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope.- This specification covers the requirements for supported vinyl dipped general purpose work gloves.

1.2 Classification.- The vinyl dipped gloves shall be furnished in the following types and sizes (see 6.2):

Type I - With gauntlet, sizes medium and large
Type II - With wristlet in one size

## 2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents.- The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

V-T-276 - Thread, Cotton
DDD-L-20 - Label; For Clothing, Equipage, and Tentage, (General Use)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing & Textile Research Facility, 21 Strathmore Road, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter:

FSC 8415

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MILITARY

MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-B-17757 - Boxes, Shipping, Fiberboard (Modular Sizes)

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods
FED-STD-595 - Colors
FED-STD-751 - Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

U. S. POSTAL SERVICE MANUAL

(Copies of the manual may be obtained for the Superintendent of Documents U. S. Government Printing Office, Washington, D. C. 20402.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues in effect on date of invitation for bids or request for proposal shall apply:

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Association, ATTN: Traffic Dept. 1616 P Street, N.W., Washington, D.C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)



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## 3. REQUIREMENTS

3.1 Guide sample.- Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample, in which case this specification shall govern.

3.2 First article.- When specified, (see 6.2) the contractor shall furnish sample unit(s) for the first article inspection and approval (see. 4.2).

3.3 Materials.-

3.3.1 Base fabric.- The base fabric for supporting the vinyl dip compounds shall be an interlock or jersey knit fabric, conforming to the following requirements:

* 3.3.1.1 Interlock knit fabric.- The interlock knit fabric shall be knit with an interlock stitch using bleached or unbleached singles carded cotton yarns. One side of the fabric shall be lightly brushed or napped to give a fleece effect. The finished fabric shall meet the requirements specified in Table I, when tested as specified in 4.3.1.

* 3.3.1.2 Jersey knit fabric.- The jersey knit fabric shall be knit with a plain jersey stitch using bleached or unbleached singles or 2 ply carded cotton yarns. The back side of the fabric shall be lightly brushed or napped to give a fleece effect. The finished fabric shall meet the requirements specified in Table I, when tested as specified in 4.3.1.

Table I - Physical requirements of knit base fabrics (minimum)

| Characteristic | Interlock knit | Jersey knit |
|---|---|---|
| Weight, ounces per square yard (g/sq.m) | 5.8 (197) | 6.0 (203) |
| Wales per inch (2.5cm) | 36 | 22 |
| Courses per inch (2.5cm) | 28 | 26 |
| Bursting strength, pounds (N) | 90 (400) | 90 (400) |

* 3.3.2 Wristlet fabric.- The fabric used for the wristlet on the type II gloves shall be a tubular (seamless) circular knit fabric knit with a 1X1 rib stitch using bleached or unbleached singles carded cotton yarns. The finished tubular fabric shall conform to the physical requirements specified in Table II, when tested as specified in 4.3.1.

Table II - Physical requirements 1X1 rib knit fabric

| Characteristic | Requirement | |
|---|---|---|
| | Minimum | Maximum |
| Weight, yards per pound (M/kg) | 12 (24) | 14 (28) |
| Wales per inch (2.5cm) | 14 | - |
| Courses per inch (2.5cm) | 20 | - |
| Flat width, inches | 3 1/4 (8.1) | - |



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3.3.3 Coating compounds.-

3.3.3.1 Base dipping compound.- The dipping compound shall be a virgin polyvinyl chloride or a vinyl chloride-acetate copolymer, containing not more than four percent polyvinyl acetate plasticized with phosphate or phthalate ester, pigmented black to approximate color chip No. 37038 of FED-STD-595. The ratio shall be a maximum of 125 parts of plasticizer to 100 parts of vinyl polymer (see 4.3.1).

3.3.3.2 Final dip coating compound.- The final dip coating shall contain unplasticized particles of powdered vinyl resin which shall impart a roughened texture to the coated gloves or as an alternate, the coated gloves may be given a chemical dip to impart a uniform roughened surface (see 4.3.1).

3.3.4 Thread.- The thread for seaming and stitching the knit support shell of the gloves shall be cotton, conforming to type IA1, ticket numbers 50 and 70, 3-ply of V-T-276.

3.3.5 Label.- Each finished type I and II glove (see NOTE:) shall be marked with a combination identification and instruction label comforming to type III or IV, classes 12 and 13, of DDD-L-20, except that the color of the marking medium shall be of any contrasting permanent-type ink and shall bear the following inscription in letters not less than 1/8 inch (3.2mm) in height:

DO NOT USE FOR ELECTRICAL WORK, BCW PROTECTION, OR FOR ROCKET FUEL OPERATIONS

GLOVES, CLOTH, VINYL DIPPED, GENERAL PURPOSE (WATER, AVIATION FUEL,
HYDROGEN PEROXIDE, OILS)
TYPE I - MEDIUM (Example)
CONTRACT NO. DSA100-00-0-000 (Example)
STOCK NO.: 8415-00-000-0000 (Example)
NAME OF CONTRACTOR:

NOTE: When the surface of the type II glove is too rough to accept a marked label a cloth label approximately 1 1/2 by 2 3/4 inches (3.8 by 7.0cm) in size and conforming to class 4 of DDD-L-20 may be used.

3.3.5.1 Instruction sheet.- A white paper instruction sheet shall be placed inside one glove of each pair with the following legend legibly printed or typed in black characters not less tha 1/8 inch (3.2mm) for title and 3/32 inch (2.4cm) for the remainder:

GLOVES, CLOTH, VINYL DIPPED, GENERAL PURPOSE (WATER, AVIATION FUEL,
HYDROGEN PEROXIDE, OILS)



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This general purpose glove is intended for wear under wet/cold operating conditions or where protection from solvents, jet fuels, oils, low concentrations of acids and other chemicals is required. It is not to be used for electrical work, BCW protection, or for rocket fuel operations. For added protection against cold/wet conditions, use a wool liner insert. To prolong life of gloves, rinse with water or wash with soap solution, especially after contact with chemicals or oils.

3.4 Physical requirements.- The finished types I and II glove shall conform to the requirements specified in table III when tested as specified in 4.4.

Table III - Physical requirements

| Characteristic | Requirements Minimum | Maximum |
|---|---|---|
| Bursting strength, pounds (N) | 50.0 (223) | - |
| Stiffness (cms) | | |
| at 21 ± 2°C in finger direction only | - | 11.0 |
| at -18 ± 2°C in finger direction only | - | 12.0 |
| Blocking | - | Scale Rating No. 2 |
| Porosity 1/ | Pass - | |
| Thickness of coating on the smooth surface (inches) | 0.020 | 0.035 |
| JP-4 fuel resistance 2/ | Pass | |

1/ There shall be no evidence of air bubbles.
2/ There shall be no evidence of cracking, softening, flaking, or separation between coating and fabric or between base and final coatings.

* 3.5 Design.- The types I and II gloves shall have a four fingers and thumb silhouette, using a two piece stitched knit fabric support shell. The type I glove shall have a grown-on gauntlet and the type II glove shall have a knitted wristlet stitched to the support shell (see figures 1 and 2).

3.5.1 Figures.- Figures 1 through 3 are furnished for information purposes only. When inconsistencies exist between the written specification and the figures, the written specification shall govern.

3.6 Dipping forms and patterns.- The dipping forms and the dies or patterns for cutting all parts of the knitted support shell shall be furnished by the contractor, and shall be of the proper proportions to provide good fitting gloves conforming to finished measurements specified in Table V and VI.



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3.7 Construction.-

3.7.1 Stitches, seams, and stitchings.- The stitch, seam, and stitching types specified in Table IV shall conform to FED-STD-751. Whenever two or more methods for seam or stitch types are given for the same part of the operation, any one of them may be used. Unless otherwise specified, all seam allowances shall finish 1/8 inch (0.3cm) minimum except when attaching the wristlet on the type II support shells. Trimmer attachments may be used. For the type II support shells, the wristlet shall be stitched to the hand portion 1/4 to 3/8 inch (0.6 to 1.0cm) from the raw edge.

3.7.2 Stitches per inch.- The range of stitches per inch as specified in Table IV shall be interpreted to mean the minimum and maximum number of stitches permitted.

3.7.3 Thread breaks and ends of seams.- Ends of all seams and stitchings produced with 301 stitch type shall be backtacked not less than 3/8 inch (1.0cm). Both ends of all seams and stitchings produced with 401 stitch type, when not caught in other seams or stitching, shall have a 1/2 to 3/4 inch (1.3 to 1.9cm) chain extending beyond each end. Thread breaks (all stitch types) shall be secured by stitching back of the break not less than 1/2 inch (1.3cm). Skipped stitches or thread breaks in 401 stitch type may be repaired using 301 stitch type.

* 3.8 Manufacturing operations requirements.- The gloves shall be made in accordance with the operations listed in Table IV. The contractor is not required to follow the exact sequence of operations listed provided that the finished gloves are identical to those produced by following the sequence of operations listed in Table IV.



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**Table IV - Construction of glove**

| No. | Description of Operation | Stitch Type | Seam and Stitching Type | Stitches Per Inch | Thread | |
|---|---|---|---|---|---|---|
| | | | | | Needle | Bobbin or Looper |
| 1. | Cutting.-<br>a. Cut the component parts of a two-piece support shell in the length (wale) direction from the base fabrics specified in 3.3.1.1 or 3.3.1.2.<br><br>b. Cut the wristlet for the type II glove from the fabric specified in 3.3.2 in sufficient length to permit formation of a double thickness wristlet that meets the required finished length measurement.<br><br>c. The unnapped side of the base fabric shall be the face side or the side to be coated. The napped side of the base fabric shall be the back side or the lining side. | | | | | |
| 2. | Replacement of defective components.-<br>During the cutting and manufacturing process, components having material defects or damages that are classified as defects in Section 4 shall be removed from production and replaced with nondefective and properly matched components. | | | | | |
| 3. | Marking.-<br>All component parts of the base material shall be marked, bundled, or ticketed to assure proper assembly. | | | | | |
| 4. | Make support shell.-<br>Join the two pieces of knit fabrics, face to face, stitching along the periphery of the side, thumb, fingers and opposite side of the shell to the top open edge in one continuous operation. | 301<br>or<br>401 | SSa-1 | 10-12 | 50/3<br>50/3 | 50/3<br>70/3 |

NCTR Form 1, 15 Sep 1967





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Table IV - Construction of glove (cont'd)

| No. | Description of Operation | Stitch Type | Seam and Stitching Type | Stitches Per Inch | Thread | |
|---|---|---|---|---|---|---|
| | | | | | Needle | Bobbin or Looper |
| 5. | Attach wristlet (Type II glove).- | | | | | |
| | a. Align raw edges of double thickness knit wristlet with top raw edge of hand portion of support shell and stitch 1/4 to 3/8 inch (0.6 to 1.0cm) from raw edges around entire circumference of wrist. | 301 or 401 or 502 thru 505 | SSa-1 | 10-12 | 50/3<br>50/3<br>50/3 | 50/3<br>70/3<br>70/3 |
| | b. When a cloth label is used it shall be simultaneously inserted and caught in the stitching of the above operation. Label shall be located on back of the right hand glove. | | | | | |
| 6. | Prepare support shell for dipping.- | | | | | |
| | a. Turn support shell so that the seam allowances are on the inside of glove as worn. | | | | | |
| | b. The support shell shall be mounted on its dipping form with the napped side facing the form and the seams at the sides and tips of fingers aligned with the corresponding locations on the dipping form. | | | | | |
| 7. | Dipping of support shell.- | | | | | |
| | a. Type I Glove - The mounted support shell shall be evenly and completely dipped in the base dipping compound specified in 3.3.3.1.<br>Type II Glove - The mounted support shell shall be evenly dipped in the base dipping compound specified in 3.3.3.1 to a point 1/2 to 1 1/2 inches (1.3 to 3.8cm) above the wristlet seam. | | | | | |
| | b. For type I and type II gloves, the rough finish imparted by the final dip compound specified in 3.3.3.2 shall extend in length from the tips of the fingers and thumb to a point 1 1/2 to 2 1/2 inches (3.8 to 6.4cm) above the thumb crotch. | | | | | |





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Table IV - Construction of glove (cont'd)

| No. | Description of Operation | Stitch Type | Seam and Stitching Type | Stitches Per Inch | Thread | |
|---|---|---|---|---|---|---|
| | | | | | Needle | Bobbin or Looper |
| 7. | Dipping of support shell.- (cont'd)<br>c. After application of final dip compound the coated gloves shall be fully cured (fused). | | | | | |
| 8. | Trimming.-<br>Type I Glove - The edge of gauntlet opening shall be evenly trimmed with a pinked or straight cut edge. | | | | | |
| 9. | Labeling.-<br>a. Type I Glove - The marking shall be applied on the outside back of the right hand glove with top edge of top line positioned approximately 3/4 inch (1.9cm) down from and parallel to top edge of glove.<br><br>b. Type II Glove - When possible, the marking shall be applied to the outside back of the right hand glove with top edge of top line positioned approximately 1/2 inch (1.3cm) below and parallel to the wrist seam. (see operation 5.b for alternate cloth label). | | | | | |
| 10. | Pairing.-<br>The finished gloves of each type shall be matched, paired, and fastened together in accordance with the best commercial practice which will permit the gloves to be separted without tearing the gloves. | | | | | |







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3.9 Measurements.- The sizes and measurements of the finished gloves shall conform to the dimensions specified in Tables V and VI. All measurements and tolerances are expressed in inches, and shall be made with the glove laid flat under no tension with the palm side face up.

Table V - Finished measurements of type I glove 1/

| Measurement | Sizes Medium | Large | Tolerance |
|---|---|---|---|
| (A) Overall length | 12 | 12 | - 1/4, + 1/2 |
| (B) Length from thumb crotch to tip of index finger | 4 5/8 | 4 5/8 | ± 1/4 |
| (C) Length of second finger | 3 | 3 | ± 1/4 |
| (D) Length of index finger | 2 1/2 | 2 1/2 | ± 1/4 |
| (E) Length of fourth finger | 2 | 2 | ± 1/4 |
| (F) Length of third finger | 2 3/4 | 2 3/4 | ± 1/4 |
| (G) Length of thumb | 2 1/2 | 2 1/2 | ± 1/4 |
| (H) Width of palm | 4 3/4 | 5 1/4 | - 1/4, + 3/8 |
| (J) Width of opening | 6 | 6 | ± 1/2 |

NOTE: Refer to figure 1 for measurements A thru J

(A) Measurement taken from tip of second finger to top edge of glove.
(B) Measurement taken from thumb crotch to tip of index finger.
(C) Measurement taken from first finger crotch to tip of second finger.
(D) Measurement taken from first finger crotch to tip of index finger.
(E) Measurement taken from third finger crotch to tip of fourth finger.
(F) Measurement taken from second finger crotch to tip of third finger.
(G) Measurement taken from thumb crotch to tip of thumb.
(H) Measurement taken across palm at base of thumb, folded edge to folded edge.
(J) Measurement taken across the top edge, folded edge to folded edge.

1/ Multiply inches by 2.5 to obtain centimeters.



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Table VI - Finished measurements of type II glove 1/

| Measurement | One Size | Tolerance |
|---|---|---|
| (A) Overall length | 10 1/4 | ± 1/4 |
| (B) Length from thumb crotch to tip of index finger | 4 5/8 | ± 1/4 |
| (C) Length of second finger | 3 | ± 1/4 |
| (D) Length of index finger | 2 1/2 | ± 1/4 |
| (E) Length of fourth finger | 2 | ± 1/4 |
| (F) Length of third finger | 2 3/4 | ± 1/4 |
| (G) Length of thumb | 2 1/2 | ± 1/4 |
| (H) Width of palm | 5 | ± 1/4 |
| (J) Width of wristlet | 3 1/2 | ± 1/4 |
| (K) Length of wristlet | 2 1/2 | ± 1/4 |

NOTE: Refer to figure 2 for measurements A thru K.

(A) Measurement taken from tip of second finger to top edge of wristlet.
(B) Measurement taken from thumb crotch to tip of index finger.
(C) Measurement taken from first finger crotch to tip of second finger.
(D) Measurement taken from first finger crotch to tip of index finger.
(E) Measurement taken from third finger crotch to tip of fourth finger.
(F) Measurement taken from second finger crotch to tip of third finger.
(G) Measurement taken from thumb crotch to tip of thumb.
(H) Measurement taken across palm at base of thumb, folded edge to folded edge.
(J) Measurement taken across the uncoated wristlet, folded edge to folded edge, approximately 1 inch down from the top edge.
(K) Measurement taken from top edge of wristlet to bottom edge at seam.

1/ Multiply inches by 2.5 to obtain centimeters.

3.10 Workmanship.- The finished gloves shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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4.1.1 Certificate of compliance.- Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 First article inspection.- The first article submitted in accordance with 3.2, shall be inspected for compliance with design, construction, workmanship and dimensional requirements.

4.3 Inspection.- Inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated herein.

* 4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document. A certificate of compliance shall be furnished for the coating compound requirements of 3.3.3.1 and 3.3.3.2. In addition, components listed in Table VII shall be tested for the characteristics specified in accordance with the referenced test methods of FED-STD-191. The physical and chemical values specified in Section 3, except where otherwise specified, apply to the average of the determinations made on the sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final results. The sample unit for test purposes shall be 1/2 yard (m) full width of each of the finished cloths. The lot size shall be expressed in units of yards (m) and shall be unacceptable if one or more sample units fail to meet any of the test requirements specified. The sample size shall be in accordance with the following:

| Lot size (yards) | Sample size (units) |
|---|---|
| 800 or less | 2 |
| 801 thru 22,000 | 3 |
| 22,001 and over | 5 |



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Table VII - Component testing

| Component | Characteristic | Requirement paragraph | Test method |
|---|---|---|---|
| Base fabric (interlock or jersey knit) | Material identification | 3.3.1.1 and 3.3.1.2 | 1200 1/ |
| | Type of knit | 3.3.1.1 and 3.3.1.2 | Visual 1/ |
| | Weight | Table I | 5041 |
| | Wales per inch | Table I | 5070 |
| | Courses per inch | Table I | 5070 |
| | Bursting strength | Table I | 5120 |
| Wristlet tubing | Material identification | 3.3.2 | 1200 1/ |
| | Type of knit | 3.3.2 | Visual 1/ |
| | Weight | Table II | 5041 |
| | Wales per inch | Table II | 5070 |
| | Courses per inch | Table II | 5070 |
| | Width (flat) | Table II | 5020 |

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

4.3.2 Examination of fabric support shell.- Examination of the completely assembled knit fabric support shells shall be made prior to dipping. The defects found shall be classified in accordance with 4.3.2.1. The sample unit shall be one support shell. The lot size shall be expressed in units of one support shell. The inspection level shall be level II of MIL-STD-105. The acceptable quality level shall be 2.5 major defects and 6.5 (major and minor defects combined) defects per hundred units.

* 4.3.2.1 Visual examination.-

| Defects | Classification Major | Minor |
|---|---|---|
| I MATERIAL DEFECTS AND WORKMANSHIP DAMAGES | | |
| a. Any hole, cut, tear, run, drop stitch, needle chew, misknit or mend | X | |
| b. Any loose knitting, causing sleaziness | | X |
| c. Pulled or snagged yarn | | X |
| d. Base fabric (interlock or jersey knit) not brushed or napped as specified | X | |



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| Defects | Classification Major | Minor |
|---|---|---|
| II CLEANNESS | | |
| a. Thread ends not trimmed or loose threads not removed | | X |
| b. Surplus seam allowance material not trimmed off trimmed unevenly or trimmed to less than 1/8 inch (0.3cm) in width | | X |
| III CUTTING | | |
| Any part not cut in accordance with specification requirements | X | |
| IV SEAMS AND STITCHINGS | | |
| a. Accuracy of stitching | | |
| 1. Any part of support shell caught in an unrelated stitching operation | X | |
| 2. Seam puckered, twisted, or pleated | X | |
| 3. Ends of stitching produced with stitch type 301 backtacked less than 3/8 inch (1.0cm) | | X |
| 4. Ends of stitching produced with stitch type 401, when not caught in other seams or stitching, having chain extending less than 1/2 inch (1.3cm) or more than 3/4 inch (1.9cm) beyond each end | | X |
| 5. Ends of stitching produced with stitch type 401, when not caught in other seams or stitching, not having chain extending beyond each end | X | |
| 6. Thread breaks stitched less than 1/2 inch (1.3cm) back of break | X | |
| b. Stitching margin | | |
| 1. Irregular, affecting serviceability or fit | X | |
| 2. Not within width(s) specified | | X |
| c. Any open seam | | |
| (One or more broken stitches or two or more continuous skipped or runoff stitches constitute an open seam) | X | |
| d. Seam and stitch types | | |
| Not specified seam or stitch type | X | |



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| Defects | Classification Major | Minor |
|---|---|---|
| IV SEAMS AND STITCHINGS (cont'd) | | |
| e. Stitch tension | | |
| 1. Loose tension resulting in a loosely secured seam | X | |
| 2. Tight tension (stitches break when normal strain is applied in the direction of the seam or stitching) | X | |
| f. Stitches per inch (to be scored only when condition exists on more than 1/2 the length of the continuous joining seam) | | |
| 1. One stitch less than the minimum or any number in excess of the maximum | | X |
| 2. Two or more stitches per inch less than the minimum specified | X | |
| V COMPONENT AND ASSEMBLY | | |
| a. Any component part or required operation omitted | X | |
| b. Any component part twisted or distorted | X | |
| c. Any required operation not as specified | | X |
| d. Any component other than type specified | | X |

4.3.3 Examination of the end item.- The defects found during the examination of the end item shall be classified in accordance with 4.3.3.1 and 4.3.3.2. The applicable inspection levels and acceptable quality levels (AQL's) shall be as indicated in 4.3.3.3. The lot size shall be expressed in terms of one finished glove. The sample unit shall be one finished glove, and the selection shall be by pairs. Defects for pairing shall be classified as a single defect.

* 4.3.3.1 Visual examination.-

| Examine | Defect | Classification Major | Minor |
|---|---|---|---|
| I PAIRING | a. Definite variation in color or appearance | | X |
| | b. Not mated properly e.g., two left, two right gloves, not same size | X | |
| | c. Pair not securely fastened together | | X |
| II CLEANNESS | Any permanent spot or stain on outside | | X |



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| Examine | Defect | Classification Major | Minor |
|---|---|---|---|
| III DESIGN | Not as specified | X | |
| IV COATED FABRIC | a. Any hole, cut or tear, thru material or exposing base fabric | X | |
| | b. Any evidence of separation of coating | X | |
| | c. Any evidence of excessive coating strike-thru on uncoated lining side | X | |
| | d. Any area of no coating | X | |
| | e. Any brittle area which when bent results in a crack | X | |
| | f. Any area of excessive coating, i.e., solid runs, ridges, or lumps | | X |
| | g. Any pitted or abraded area: | | |
| | 1. Base fabric not exposed | | X |
| | 2. Base fabric exposed | X | |
| | h. Any imbedded foreign material which when removed will expose base fabric or leave a hole thru material | X | |
| | i. Rough final coat omitted | X | |
| | j. Rough final coat does not extend 1 1/2 to 2 1/2 inches (3.8 to 6.3cm) above thumb crotch | | X |
| | k. Color does not approximate specified color chip | | X |
| | l. Any tackiness | X | |
| | m. Any blister | X | |
| | n. Any imbedded foreign matter: | | |
| | 1. -affecting serviceability | X | |
| | 2. -affecting appearance | | X |
| | o. For type II glove only: coating on knit wristlet extends more than 1 1/2 inches (3.8cm) beyond joining seam. | | X |
| V WORKMANSHIP | Any evidence that seams in support shell were not properly aligned with those on dipping form | | X |
| VI LABEL | a. Marking or label omitted, incorrect or illegible | X | |
| | b. Misplaced | | X |
| | c. Marking not permanent, rubs off | X | |
| | d. Label not securely caught in seam | | X |



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4.3.3.2 Dimensional examination.- The gloves shall be examined for defects in dimensions as specified in tables V and VI. Any measurement deviating from nominal dimensions and tolerances specified shall be scored as a measurement defect.

4.3.3.3 Inspection levels and acceptable quality levels (AQL's).- The inspection levels and acceptable quality levels, expressed in defects per 100 units for visual and dimensional examination shall be as follows:

| | AQL's | | |
|---|---|---|---|
| | Major | Total | Inspection level |
| For visual examination in 4.3.3.1 | 2.5 | 6.5 | II |
| For dimensional examination in 4.3.3.2 | - | 4.0 | S-3 |

4.3.4 Examination of packaging requirements.- An examination shall be made to determine that packaging, packing, and marking complies with Section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery, the exception that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per 100 units.

| Examine | Defect |
|---|---|
| Markings (exterior and interior) | Omitted, incorrect, illegible, of improper size, location, sequence or method of application |
| Materials | Any component missing, damaged, or not as specified. |
| Workmanship | Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, inadequate stapling or improper taping. Bulging or distortion of containers. |
| Content | Number of pairs per container is more or less than required. Number of pairs per bundle is more or less than required. 1/ |

1/ For this defect, one bundle shall be examined from each shipping container in the sample.

4.3.2 Examination for palletization (Type II Gloves only).- When palletization is specified, an examination shall be made to determine that the palletization complies with the Section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully prepared for delivery. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL shall be 6.5 defects per 100 units.



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| Examine | Defect |
|---|---|
| Finished dimension | Length, width, or height exceeds specified maximum requirement. |
| Palletization | Not as specified. Pallet pattern no as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified. |
| Weight | Exceeds maximum load limits. |
| Marking | Omitted, incorrect, illegible, of improper size, location, sequence or method of application. |

* 4.4 Testing the end item.- Each lot of the end item shall be tested for the characteristics shown in Table VIII and as specified in FED-STD-191, whenever applicable. Unless otherwise specified, the sample unit for tests shall be nine(9) pairs of type I gloves and four (4) pairs of type II gloves. The lot size shall be expressed in units of one glove each, and the selection shall be made in pairs. The inspection level shall be S-1 with an AQL of 4.0 defects per 100 units, except that any failure excountered in the porosity test or the JP-4 fuel resistance test shall cause rejection of the lot.

Table VIII - End item tests

| Characteristic | Requirement paragraph | Test method |
|---|---|---|
| Bursting strength | Table III | 5120 |
| Stiffness | Table III | 5204 1/ |
| Thickness | Table III | 5030 2/ |
| Blocking | Table III | 5872 3/ |
| Porosity | Table III | 4.4.1 |
| JP-4 fuel resistance | Table III | 4.4.2 |

1/ For the type II gloves, the test specimen shall be cut in a straight line parallel to the finger direction and into the long fingers. Requirement that the specimen be selected from material which has not been creased or folded in any manner is not applicable.

2/ Using the coated cloth method measure the thickness of the smooth dipped area in three different areas. Strip a sufficient amount of coating from the base fabric for each individual thickness reading by either the dry peel method or wet peel method (soaked in methyl ethyl ketone). Measure thickness of base fabric where coating has been peeled off. The difference between base fabric and coated fabric shall be the coating thickness.

3/ Except that the test specimen shall be 2x4 inches (5x10cm), the glass plates shall be approximately 3x3 inches (7.6 x 7.6cm), and the weight shall be 2 pounds (0.9kg).



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4.4.1 Porosity.-

4.4.1.1 Apparatus.- The porosity test apparatus shall consist of a hollow wooden or metal shank, 3 inches long, tapered on the outer surface to fit the open end of the glove; and a steel ring tapered on the inner surface to fit over the wooden shank. The shank shall be attached to a steel base plate to make an airtight joint, and the tapered ring shall be bolted to the base plate so that it can be tightened over the shank. The base plate shall be fitted with a 1/4 inch (0.6cm) internal pipe size brass nipple which shall be connected with other pipe fittings to mount a tire inflation valve, an air pressure gage, and a pressure release valve (see figure 3). Any type apparatus similar in principle to that specified herein may be used.

4.4.1.2 Procedure.- The opening of the glove shall be pulled over the shank of the test apparatus and clamped firmly in place by tightening the steel ring against the base plate. Compressed air shall be fed into the glove in sufficient quantity to fill out the glove and to provide a slight additional pressure. The inflated glove shall be immersed in water for 1 minute for observation of porosity as indicated by air bubbles. No visible bubbles shall be permitted within the 1 minute period. One determination shall be made per sample unit and the results reported as "pass" or "fail".

4.4.2 JP-4 fuel resistance test.- A test specimen 3x3 inches (7.6x7.6cm) shall be cut from the rough surface portion of the glove. The specimen shall be immersed completely for 15 minutes in JP-4 fuel, conforming to MIL-T-5624. Remove the specimen and allow to dry at room temperature for four hours. When dry, the test specimen shall be folded in half and rolled with a roller of approximately 10 pounds (4.5kg) in weight. The test specimen shall be folded again in half, perpendicular to the direction of the first fold, and rolled again in the same manner. The specimen shall be examined for evidence of cracking, softening, flaking, or separation of coating from the fabric. One determination shall be made per sample unit and the results reported as "pass" or "fail".

5. PACKAGING

5.1 Preservation-packaging.- Preservation-packaging shall be level A or C as specified (6.2).

5.1.1 Level A.-

5.1.1.1 Type I.- Twelve pairs of type I gloves of one size only shall be evenly stacked with each pair alternately reversed to form a neat bundle measuring approximately 15 inches (38cm) in length. The bundles shall be securely tied approximately three inches (7.6cm) from each end with cotton tape or twine.



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5.1.1.2 Type II.- Eight pairs of type II gloves shall be evenly stacked with each pair alternately reversed to form a neat bundle. The bundle shall be securely tied with cotton tape or twine.

5.1.2 Level C (Commercial packaging).- Gloves shall be packaged to afford adequate protection against physical damage during shipment from the contractor to the first receiving activity. The package and quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.- Forty-eight pairs of gloves of one type and size only and packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed, and reinforced conforming to type SF class weather-resistant, grade V2s, size 2A of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class weather-resistant, variety DW, grade V15c of MIL-B-17757. The type I gloves shall be packed flat, four in length with sides of gloves parallel to the ends of the container, one in width, and one in depth within a shipping container. The type II gloves shall be packed two in length with sides of gloves parallel to length of container, three in width, and one in depth within a shipping container. Voids within the container shall be filled with cushioning material to immobilize the type II gloves. Each container shall have the contents completly covered on the top and bottom with a sheet of any suitable grade of commercial kraft paper.

5.2.2 Level B.- Forty-eight pairs of gloves of one type and size only and packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled and closed conforming to type CF class domestic, grade 275, size 2A of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class domestic, variety DW, grade 200 of MIL-B-17757. The type I gloves shall be packed flat four in length with sides of gloves parallel to ends of container, one in width, and one in depth within a shipping container. The type II gloves shall be packed flat two in length with sides of gloves parallel to length of container, three in width, and one in depth within a shipping container. Voids within the container shall be filled with cushioning material to immobilize the type II gloves. Each container shall have the contents completly covered on the top and bottom with a sheet of any suitable grade of commercial kraft paper.

5.2.2.1 Weather-resistant fiberboard containers.- When specified (see 6.2), the shipping container shall be a grade V3c, W5c, or W6c fiberboard box fabricated in accordance with MIL-B-17757 and closed in accordance with the appendix of the box specification.

5.2.3 Level C (Commercial packing).- Gloves, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with the U.S. Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.



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5.3 Palletization.- When specified (see 6.2), the type II gloves, packed as specifed in 5.2, shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means of K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.4 Marking.- In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The gloves covered by the specification will be used by military personnel as a general purpose utility item. The gloves are designed for protection under the following conditions and operations:

(a) In the handling of low concentrations of chemical (e.g., acids, alkalis, petroleum products, paints, thinners, solvents, and degreasing materials).
(b) In cold/wet weather.
(c) During jet fueling operations.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and size required (see 1.2).
(c) Whether first article sample is required (see 3.2).
(d) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(e) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
(f) When palletization is required (see 5.3).

6.3 Samples.- For access to samples, address the procuring activity issuing the invitation for bids.

6.4. Changes from previous issue.- The margins of this specification are marked with an asterisk to indicate where changes (addition, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.



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Navy - SA
Air Force - 99

Review activities:
Navy - SH
Air Force - 11

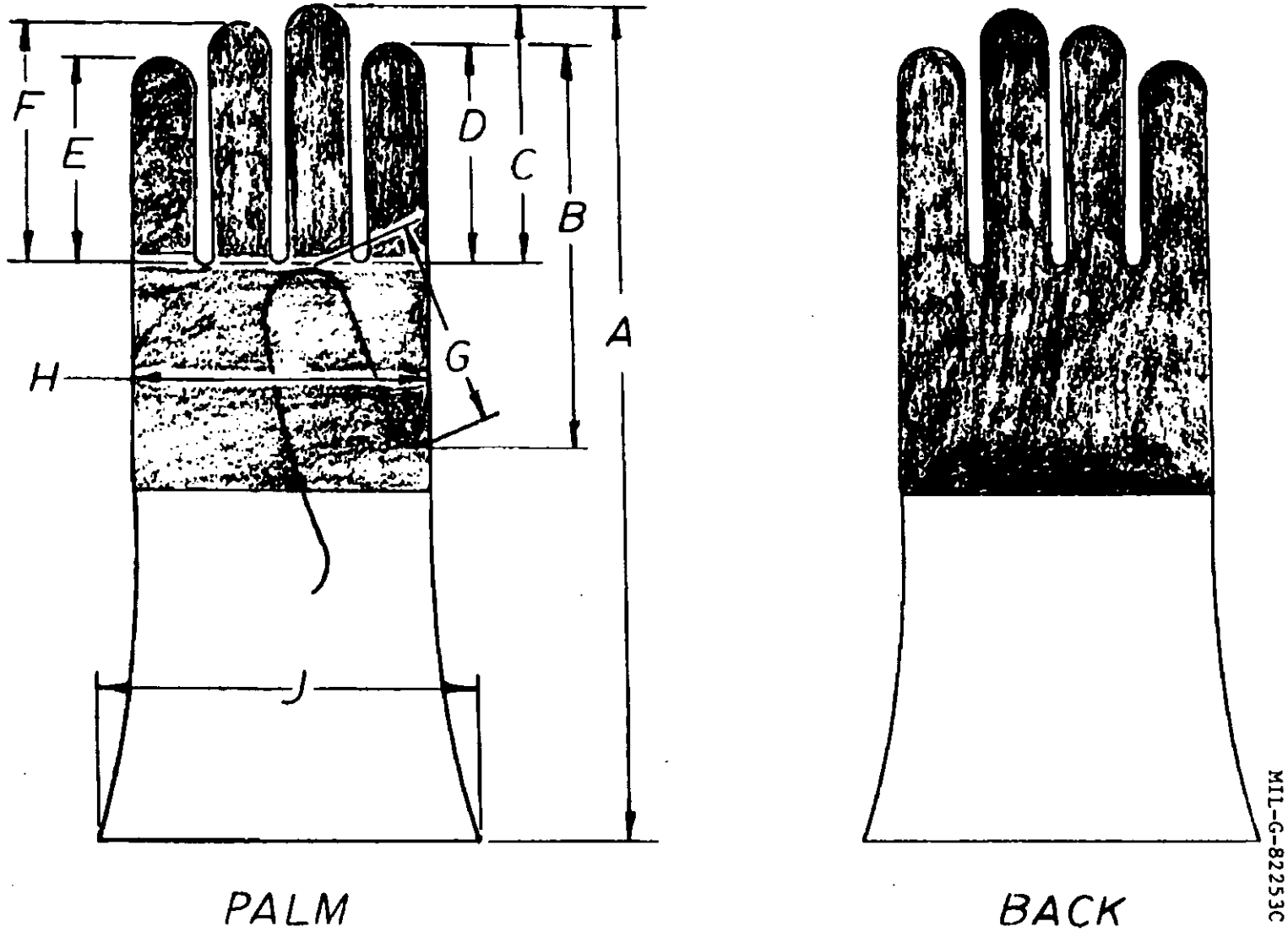
User activities:
Navy - YD, AS, MC, OS

Preparing activity:
Navy - SA

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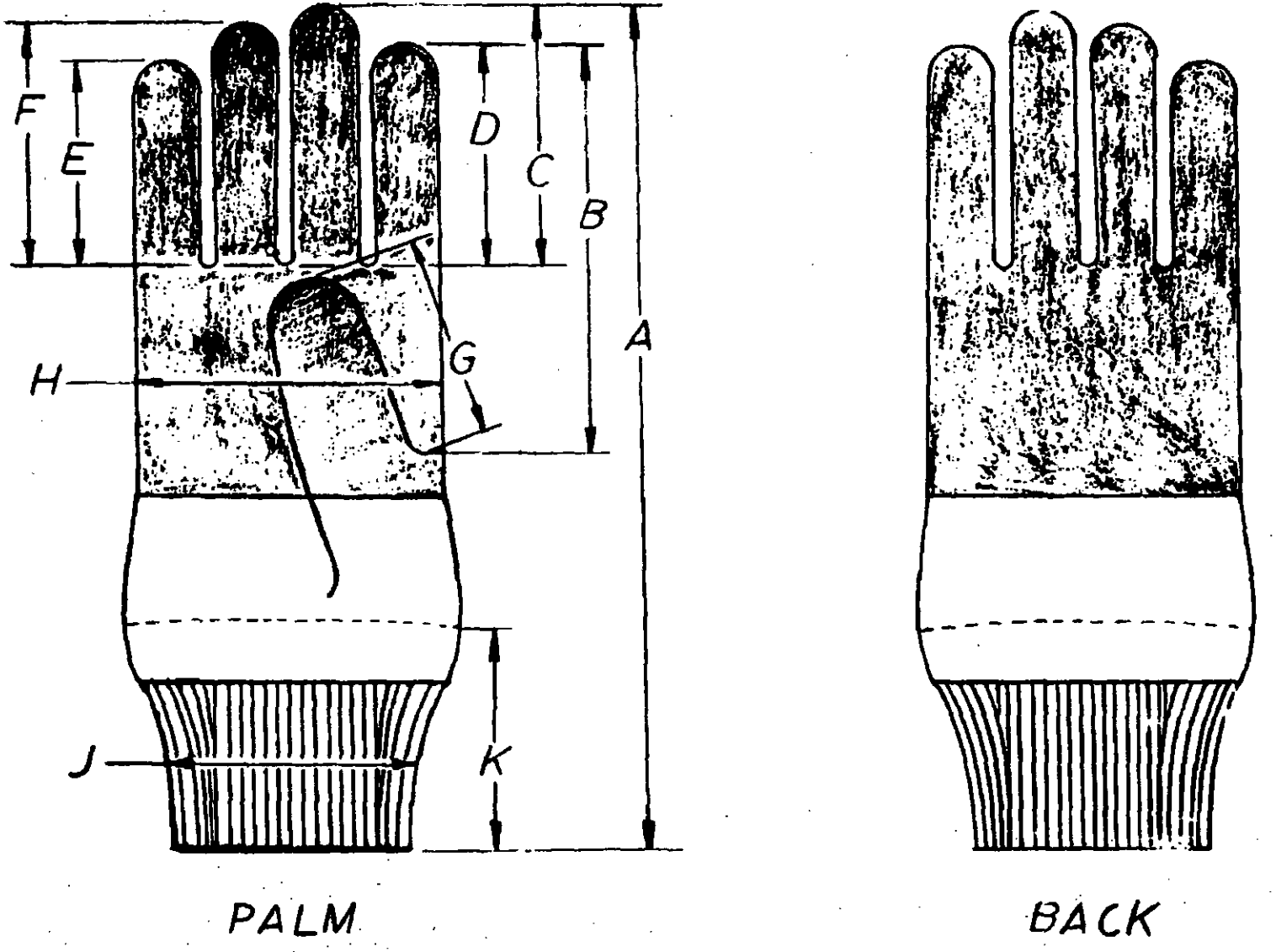



TYPE I

FIG 1 - GLOVES, VINYL DIPPED, GENERAL PURPOSE (WATER, AVIATION FUEL, HYDROGEN PEROXIDE, OILS)



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TYPE II

FIG 2 - GLOVES, VINYL DIPPED, GENERAL PURPOSE (WATER, AVIATION FUEL, HYDROGEN PEROXIDE, OILS)



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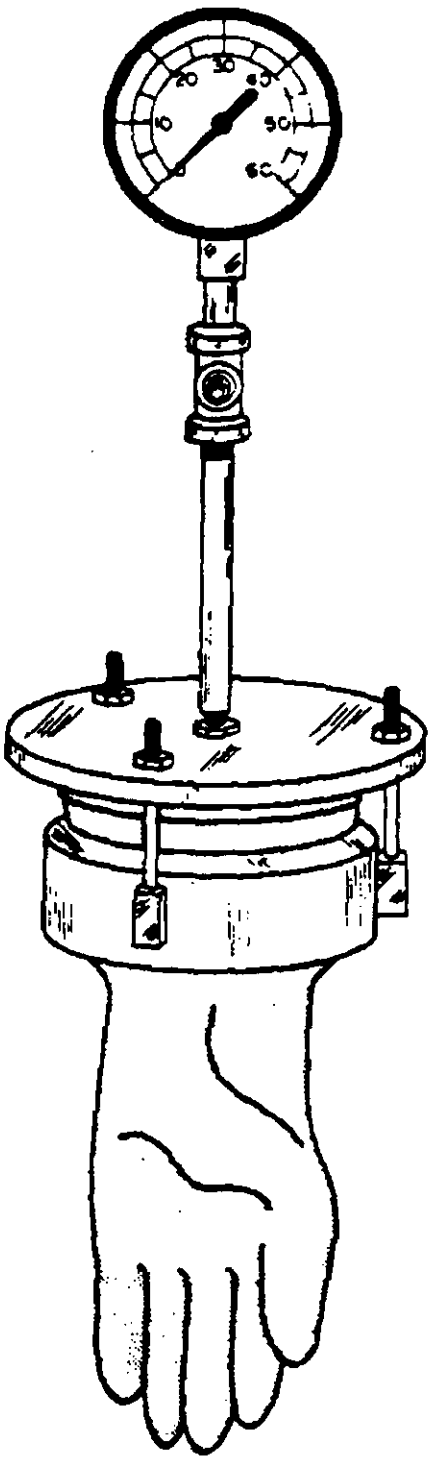



FIGURE 3 - Apparatus for porosity test

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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